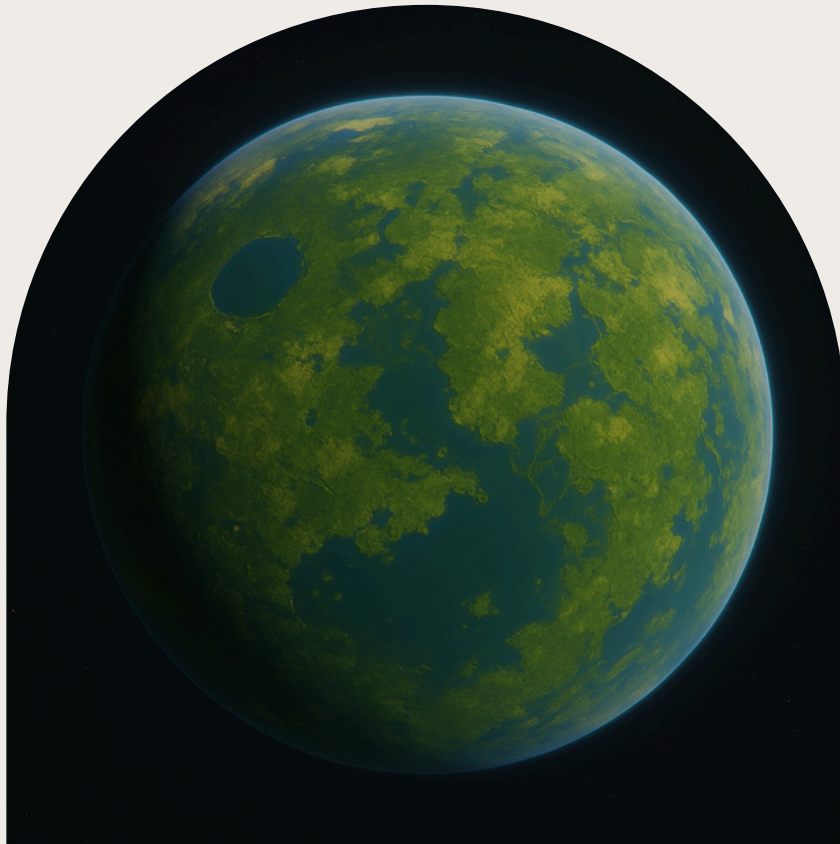


DESERTA VENTURES

THE FINAL FRONTIER



TERRAFORM MARS

Lorem ipsum

By: Jose Subiabre Abalos

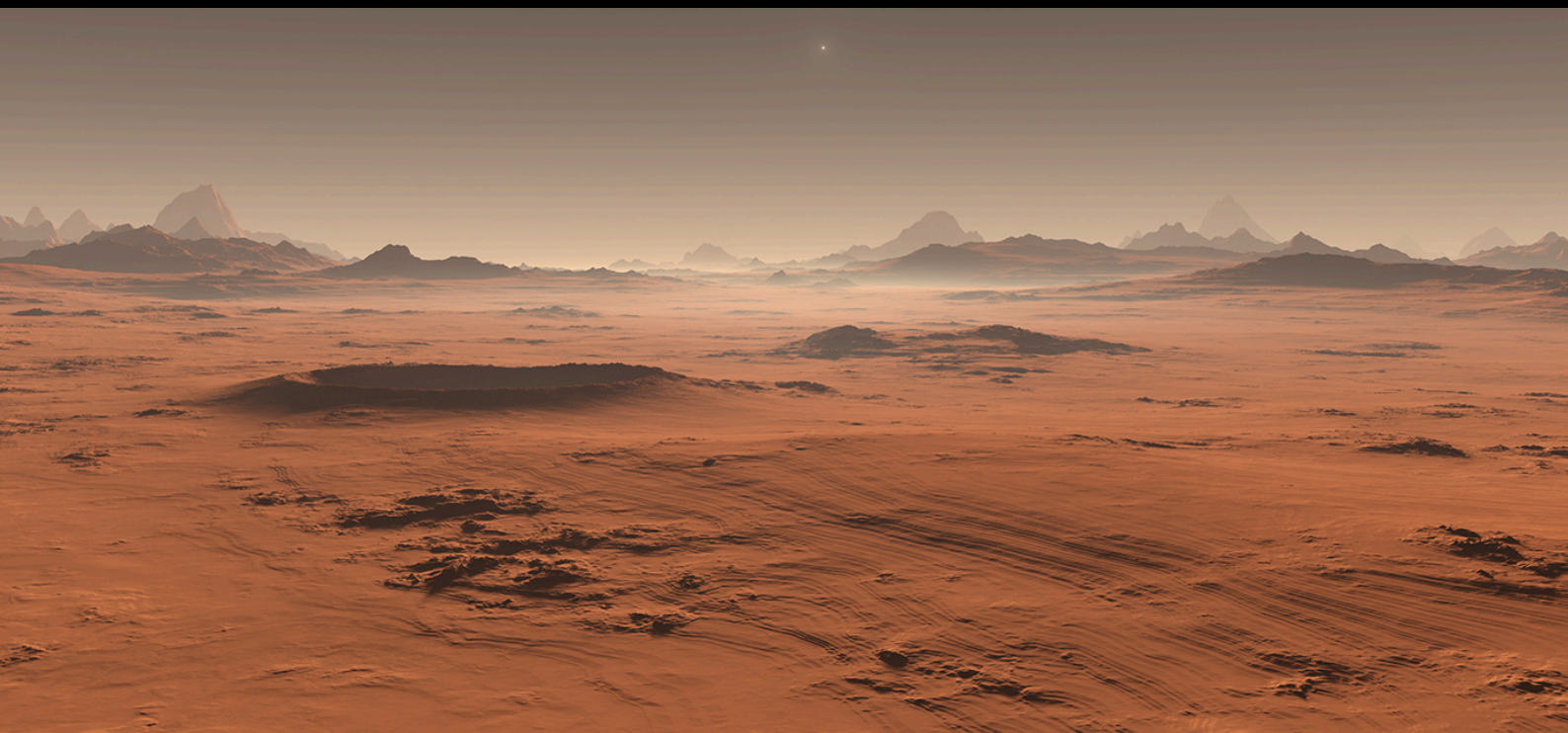
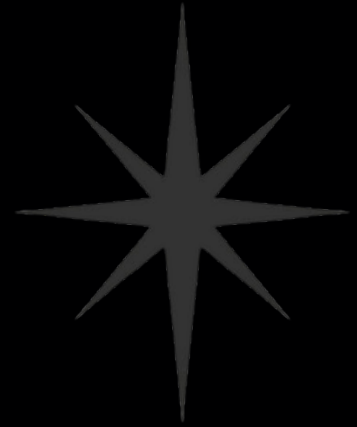
INTRODUCTION

Reversing Mars' situation—terraforming it into a blue or green world again—is theoretically possible, but not feasible with current technology. The core challenges lie in restoring its atmosphere, warmth, water, and protection from solar radiation.

Working on desertification here on Earth is one of the best real-world laboratories for developing the technologies, strategies, and ethics that could one day support terraforming Mars.

Desert regeneration work on Earth is not only applicable to Mars, it's essential. It serves as a technological testbed, a biological sandbox, and a moral rehearsal for reshaping entire ecosystems.

If we can't fix deserts here, we're not ready to build life elsewhere.

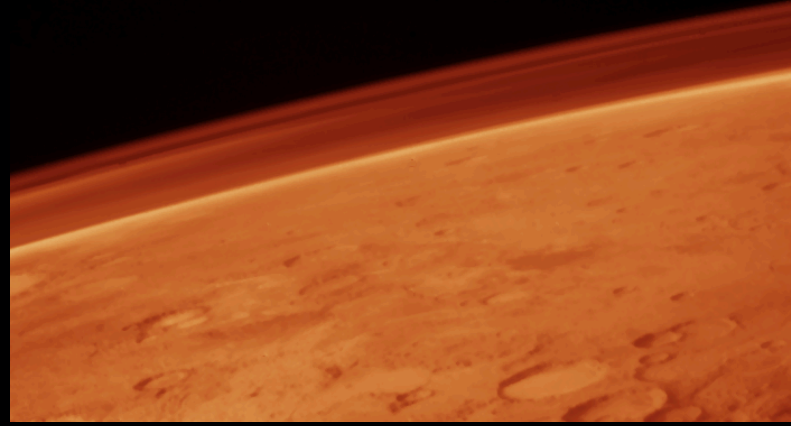


ATMOSPHERE

MA passive water collection method that uses vertical mesh nets to capture tiny water droplets from fog. As fog passes through the mesh, droplets condense, coalesce, and drip into collection troughs.

- Yield: Typically 3–10 liters per m² of mesh per day.
- Best used in: Coastal deserts (e.g., Atacama, Namib) with consistent fog and low wind.

Best for: Subtropical, Coastal, Continental Deserts



TEMPERATURE

The process of removing salt and impurities from seawater or brackish water using solar energy. It can be thermal (evaporation-condensation) or photovoltaic-powered (driving reverse osmosis pumps).

- Types: Solar stills, multi-effect distillation, or PV + reverse osmosis.
- Use case: Off-grid freshwater production in sunny, arid regions near coasts or saline aquifers.

Best for: Subtropical, Coastal, Continental Deserts

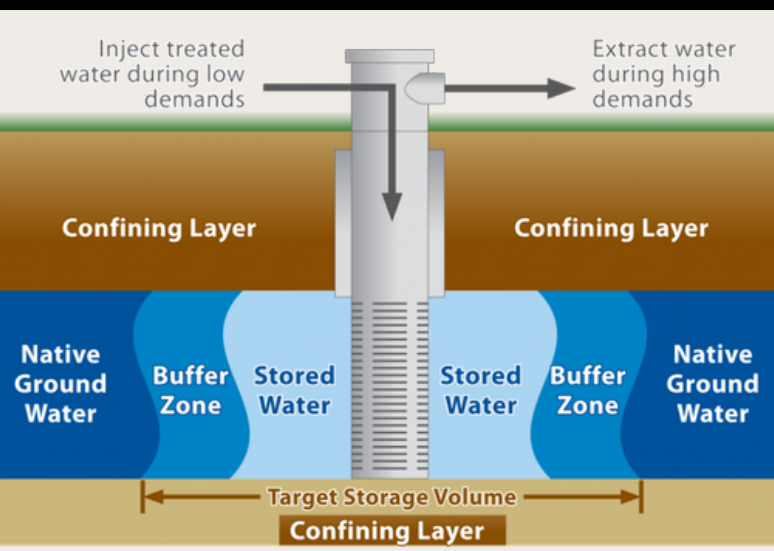


WATER

Devices that extract moisture from ambient air by cooling it below the dew point, causing condensation.

- Tech: Refrigeration-based or desiccant-based systems.
- Energy use: High (0,3–1 kWh per liter), unless paired with renewables.
- Best in: Humid environments (>40% RH); limited in dry desert interiors unless fog is present.

Best for: Coastal



RADIATION

A method to deliberately infiltrate surface water (rain, treated wastewater, floodwater) into underground aquifers to replenish groundwater.

- Techniques: Recharge basins, percolation tanks, injection wells.
- Benefit: Stores water underground to reduce evaporation and stabilize supply during dry seasons.

Best for: Subtropical, Continental Deserts

ATMOSPHERE

Why it matters: Mars has ~0,6% of Earth's atmospheric pressure—too thin to support liquid water.

Theoretical solutions:

Release CO₂ trapped in Martian soil and polar ice caps (using giant mirrors or nukes to heat them).

Import greenhouse gases like CFCs to warm the planet (though this would require thousands of factories).

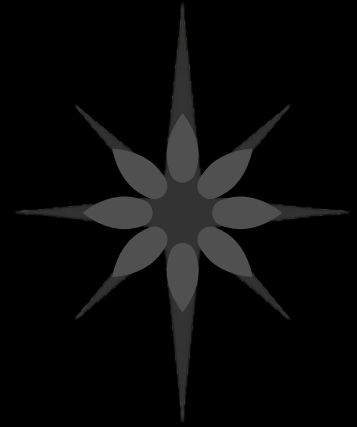
Build artificial magnetic shields at Mars' L1 point to reduce solar stripping and retain gases.

Problem: NASA and ESA studies show there isn't enough accessible CO₂ to thicken the atmosphere even if we vaporized it all.



TEMPERATURE

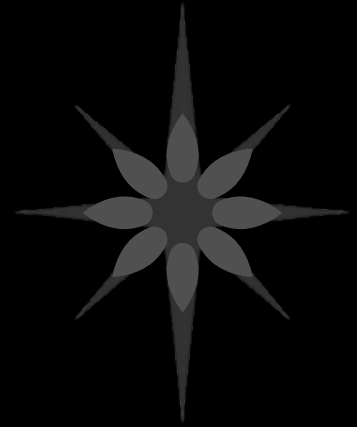
- Goal: Warm Mars to above 0°C to allow liquid water to flow.
- Proposed methods:
- Giant orbital mirrors to focus sunlight.
- Greenhouse gas production on an industrial scale.
- Black dust on ice caps to increase solar absorption.
- Timeframe: Thousands of years, even with ideal technology.



WATER

- Mars has frozen water at the poles and underground.
- Once the atmosphere is thick and warm enough, water could potentially melt and flow—re-creating rivers, lakes, and even shallow seas.

But without a strong atmosphere or pressure, any liquid water today evaporates or freezes immediately.



CAN WE TURN DESERTS INTO FORESTS?

The idea of transforming deserts into green, forested ecosystems is no longer science fiction. Projects in China, the Middle East, and Africa have shown that desert greening is technically possible, especially when done strategically. However, it requires enormous planning, infrastructure, and an understanding of the ecological limits.

Who's Doing It?

- China's Kubuqi Desert: Transformed 6,000 km² through afforestation and solar-agriculture synergy.
- Saudi Arabia's Green Initiative: Aims to plant 10 billion trees over decades, tied to Vision 2030.
- Great Green Wall (Africa): Ambitious effort to reforest across the Sahel to halt desert expansion.
- Sahara Forest Project: Combines solar energy, greenhouses, and saltwater agriculture in Jordan and Qatar.

Yes, deserts can become forests—but only where it makes ecological and economic sense. The smarter goal is not to copy rainforests, but to design climate-adapted, regenerative landscapes that bring life, food, and stability to arid zones. Forests are possible—but forests that fit the desert, not fight it.





What Makes It Possible?



Water Access

- Without water, there is no forest. Solutions include:
 - Solar-powered desalination
 - Treated wastewater
 - Atmospheric water harvesting
 - Managed aquifers

Soil Regeneration

- Desert soils are often salty, compacted, or nutrient-poor. Techniques like biochar, composting, and microbial seeding can slowly restore fertility.

Climate-Resilient Species

- Not all forests need to be tropical. Planting native, drought-resistant trees (e.g., Acacia, Prosopis, date palms) creates microclimates that allow biodiversity to follow.

Integrated Systems

- Agroforestry, agrovoltatics, and regenerative agriculture offer forest-like cover with economic outputs—food, biomass, and carbon capture—without needing high water inputs.

Technology & Policy

- Smart irrigation (drip, fog, or underground), remote sensing, and AI-driven land monitoring make it scalable. Governments must back it with land use reform, long-term funding, and legal clarity.

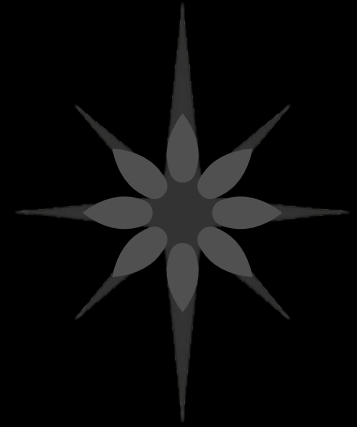
DREAMING GREEN: SAUDI ARABIA AMBITIOUS PROJECT

Saudi Green Initiative (SGI) is an ambitious initiative focused on combating climate change, improving the quality of life, and protecting the environment for future generations.

Launched in 2021 by His Royal Highness Prince Mohammed bin Salman bin Abdulaziz, Crown Prince and Prime Minister, SGI is a whole-of-society initiative, uniting all sustainability efforts in the country to rapidly scale up the Kingdom's climate action.

SGI supports Saudi Arabia's ambition to reach net-zero emissions by 2060 through the Circular Carbon Economy approach and is also accelerating the country's transition to a green economy. Three overarching targets guide SGI's work - emissions reduction, afforestation, and land and sea protection.

Since SGI's inauguration, 77 initiatives corresponding to the three targets have been activated, representing an investment of more than \$186bn and helping to drive sustainable growth. By bringing together the government and private sector and fostering collaboration and innovation, the Kingdom's commitments have been turned into tangible action while supporting global goals.



A STRATEGIC FRAMEWORK FOR DESERT REGENERATION

At Deserta Ventures, we don't believe in one-size-fits-all solutions. Every desert is different—and that matters. Building on the research from my thesis, we've developed a site-adaptable framework for desert regeneration, tailored to the specific conditions of the world's four major desert types: subtropical, cold (continental), coastal, and polar.

Whether it's fog-harvesting systems in coastal deserts, passive water capture in cold continental zones, or shade-optimized agrovoltaics in subtropical regions, our approach is flexible by design—but grounded in science and regional logic.

What we offer is a methodology—a climate-responsive system that overlays desert typologies with regenerative infrastructure. We combine solar energy, intelligent water management, and agroecological principles to design modular, scalable units that turn degraded land into living, productive ecosystems.

This isn't a dream. It's a platform built for real-world conditions—with design DNA, technical rigor, and economic logic at its core.

